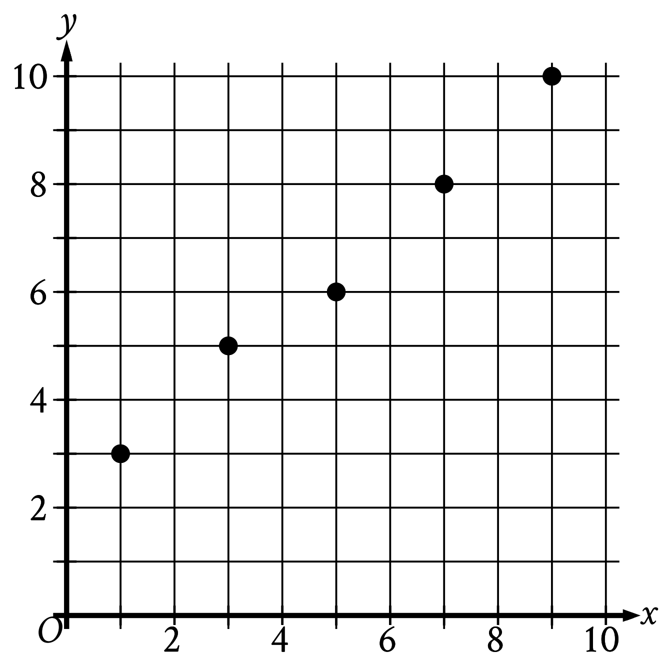
## Question ID: 16988f9c

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Easy |

**Question**

The scatterplot shows the relationship between two variables, $x$ and $y$.




Which equation is the most appropriate linear model for this relationship?

**Answer**

A. $y = -0.9x - 2.2$

B. $y = -0.9x + 2.2$

C. $y = -0.9x$

D. $y = 0.9x + 2.2$

**Correct Answer: D**

**Rationale**

Choice D is correct. A linear model can be written in the form $y = mx + b$, where $m$ is the slope of the graph of the model in the *xy*-plane and $(0, b)$ is the *y*-intercept. The graph of an appropriate linear model for this relationship passes near the points $(1, 3)$ and $(9, 10)$ in the *xy*-plane. Two points on a line, $(x_1, y_1)$ and $(x_2, y_2)$, can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(1, 3)$ and $(9, 10)$ for $(x_1, y_1)$ and $(x_2, y_2)$, respectively, in the slope formula yields $m = \frac{10-3}{9-1}$, or $m = 0.875$. Therefore, the value of $m$ for an appropriate linear model is approximately $0.875$. Substituting $0.875$ for $m$ in $y = mx + b$ yields $y = 0.875x + b$. Since an appropriate linear model passes near the point $(1, 3)$, the approximate value of $b$ can be found by substituting $1$ for $x$ and $3$ for $y$ in the equation $y = 0.875x + b$, which yields $3 = (0.875)(1) + b$, or $3 = 0.875 + b$. Subtracting $0.875$ from both sides of this equation yields $2.125 = b$. Therefore, the value of $b$ for an appropriate linear model is approximately $2.125$. Thus, of the given choices, $y = 0.9x + 2.2$ is the most appropriate linear model for this relationship.

Alternate approach: A linear model can be written in the form $y = mx + b$, where $m$ is the slope of the graph of the model in the *xy*-plane and $(0, b)$ is the *y*-intercept. The scatterplot shows that as the *x*-values of the data points increase, the *y*-values of the data points increase, which means the graph of an appropriate linear model has a positive slope. Of the given choices, $y = 0.9x + 2.2$ is the only linear model whose graph has a positive slope.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice C is incorrect. The graph of this model has a negative slope, not a positive slope.